

```

    extendedKeyUsage Extension({extnID id-ce-extKeyUsage}) OPTIONAL,
    authorityInformationAccess Extension({extnID id-pe-authorityInfoAccess}) OPTIONAL,
    subjectAlternateName Extension({extnID id-ce-subjectAltName}) OPTIONAL
}

```

The DAC certificate SHALL follow the following constraints layered on top of the encoding specified by [RFC 5280](#) within the [TBSCertificate](#) structure:

1. The [version](#) field SHALL be set to [2](#) to indicate v3 certificate.
2. The [signature](#) field SHALL contain the identifier for signatureAlgorithm [ecdsa-with-SHA256](#).
3. The [issuer](#) field SHALL be a sequence of [RelativeDistinguishedName](#) s.
4. The [issuer](#) field SHALL have exactly one [VendorID](#) value present.
5. The [issuer](#) field SHALL have exactly zero or one [ProductID](#) value present.
6. The [issuer](#) field SHALL match, byte-for-byte, the [subject](#) field of the PAI certificate for the PAI that issued this DAC.
7. The [subject](#) field SHALL be a sequence of [RelativeDistinguishedName](#) s.
8. The [subject](#) field SHALL have exactly one [VendorID](#) value present.
 - a. The VendorID value present in the [issuer](#) field SHALL match the VendorID value found in [subject](#) field.
9. The [subject](#) field SHALL have exactly one [ProductID](#) value present.
 - a. If a ProductID value was present in the [issuer](#) field, the ProductID value found in [subject](#) field SHALL match the value found in the [issuer](#) field.
10. The algorithm field in [subjectPublicKeyInfo](#) field SHALL be the object identifier for [prime256v1](#).
11. The certificate SHALL carry the following Extensions:
 - a. Basic Constraint extension SHALL be marked [critical](#) and have the [cA](#) field set to FALSE.
 - b. Key Usage extension SHALL be marked [critical](#)
 - i. The [KeyUsage](#) bitstring SHALL only have the [digitalSignature](#) bit set.
 - ii. Other bits SHALL NOT be set
 - c. Authority Key Identifier
 - d. Subject Key Identifier
12. The certificate MAY also carry the following additional Extensions:
 - a. Extended Key Usage
 - b. Authority Information Access
 - c. Subject Alternate Name
 - d. CRLDistributionPoints
 - e. Any other extension allowed in [RFC 5280](#) where inclusion does not violate size limitations. These extensions insofar not defined in this specification SHALL be ignored by commissioners conforming to this specification, but MAY be present to allow flexibility in CA operation.